



Tower Simulation System Enhancement 2 Statement of Objectives

1. Purpose

This Statement of Objectives (SOO) describes the Federal Aviation Administration's (FAA) goals and expectations for the implementation and operation of the Tower Simulation System (TSS) Enhancement 2 (E2) System.

2. Background

The Federal Aviation Administration (FAA) operates a complex air traffic control system that relies on highly trained controllers to manage aircraft movements safely and efficiently. The FAA uses high-fidelity air traffic control simulators, called TSSs, to train and evaluate controllers in performing air traffic control operations. TSSs enable the FAA to design training scenarios to meet the FAA's training objectives and allow controllers to practice air traffic control procedures in a highly realistic virtual environment without risk to National Airspace System (NAS) operations. The FAA currently operates 111 TSSs to meet the training needs of controllers stationed at all 264 FAA-operated air traffic control towers (ATCTs) throughout the NAS.

As the FAA embarks on the implementation of the Brand-New Air Traffic Control System (BNATCS) plan, the agency has an urgent need to expand the controller workforce to address current and anticipated staffing shortages. The TSS, which can accelerate controller training and reduce the time required for certification, will play an important role in helping the FAA achieve its controller staffing goals. However, there are 173 ATCTs in the NAS that do not have a TSS on site. Controllers at these ATCTs must travel to other facilities for simulation training, incurring costs and requiring time for travel that exacerbates existing controller staffing shortages. The FAA intends to acquire and deploy additional TSSs to training sites around the NAS to improve controllers' access to high-fidelity simulators and facilitate faster, more efficient controller training and certification. At the same time, the FAA will address the mandate in the FAA Reauthorization Act of 2024 ("Reauthorization Act") to improve controllers' access to high-fidelity simulation training by deploying additional TSSs in the NAS by May 2028 (PL 118-63, § 415).

3. Current Environment

The current TSS consists of hardware and software that can be configured to simulate the experience of an air traffic controller at any FAA-operated ATCT in the NAS. The TSS's consoles, headsets, surveillance, weather, and air traffic information displays, and other hardware physically and functionally emulate the air traffic control workstations in a tower cab. High-resolution monitors display realistic out-the-window views from towers based on geo-specific virtual airport models called visual databases (VDBs), which are created by the TSS vendor.

The TSS System includes software tools that enable the TSS vendor to develop training scenarios based on requirements provided by an instructor at an ATCT. The System's features and options make it possible to create training scenarios that depict a wide range of operational conditions and events that a controller could encounter. Training scenarios can also depict the particular models of aircraft and ground vehicles active at an airport. The TSS vendor maintains a digital library of moving models specified by the FAA.

The TSS can accommodate training for a single student or for multiple students practicing coordinated procedures. During training, an instructor exercises control over events, conditions, and actions in training scenarios and may also play the role of pilot for simulated aircraft while instructing and evaluating student participants. The TSS can accommodate four additional remote pilot operators (RPOs), who participate in training as additional pilots for simulated aircraft and operators for simulated ground vehicles.

The TSS vendor provides on-site operational and maintenance support for each TSS. The vendor's on-site personnel are responsible for providing training on how to operate the TSS, on-call assistance, and system troubleshooting, installing software updates, and replacing hardware when required. They are also responsible for creating and modifying training scenarios at the request of FAA-designated instructors. The vendor maintains site-specific VDBs and training scenarios at each TSS training site. The vendor also maintains a library of all VDBs at the vendor's facility. The current TSS does not have a network connection, so the vendor uses portable storage devices to transfer data and software updates between TSS sites and the vendor's facility.

The current TSS software contract expires in May 2028, and the current operational and maintenance support contract expires in September 2028.

4. Objectives

The FAA intends to deploy TSSs to 102 training sites around the NAS to improve controllers' access to high-fidelity simulators and facilitate faster, more efficient controller training and certification. The FAA is also seeking to improve the capabilities

of the TSS and the delivery of TSS services through technological innovation, including artificial intelligence. The FAA has designated this effort “TSS E2.”

The objectives of TSS E2 are as follows:

1. **High-fidelity simulation.** Enable the FAA to deliver high-fidelity air traffic control simulation training for any FAA-operated ATCT at designated TSS sites in accordance with the *TSS System Specification* (attached).
2. **Realistic virtual pilots.** Eliminate the need for RPOs to participate in training events by implementing highly realistic virtual pilots for simulated aircraft. The TSS’s virtual pilots must accurately and reliably translate student speech into text, recognize spoken commands, and effect an appropriate response to spoken commands in training scenarios in a synthetic voice that emulates natural human speech.
3. **Digital repository.** Create and maintain an online digital repository of the FAA’s VDBs. A user must be able to upload VDBs and training scenarios from any TSS to the repository and download VDBs and training scenarios from the repository to any TSS.
4. **Remote instructor participation.** Enable TSS instructors to perform the role of Supervisor from any remote location and have the same capabilities from a remote location as from the student’s location.
5. **Automated student evaluation.** Provide automated, objective evaluation of and feedback on student performance during training scenarios.
6. **Software Design and Architecture**
 - Use open standards for visual databases and training scenarios to facilitate reusability by other software applications.
 - Use modular architecture that facilitates updates and changes to the solution software.
 - Use an expandable architecture that supports incremental inclusion of new training functionality.
 - Use a modern, scalable architecture that operates reliably across diverse hosting environments, prioritizing consistency, portability, and efficiency while supporting flexible deployment in an agile and dynamic IT landscape.
 - Integrate with existing and future FAA systems, services, infrastructure, and programs.

- Prioritize resiliency to minimize downtime, withstand failures, and accelerate restoration of service.
7. **Performance reporting.** Provide the FAA with real-time access to usage data for each simulator, scenario, and simulated airport by location to enable the FAA to monitor and evaluate the performance of the TSS solution.
 8. **Integrated logistics support.** Provide the FAA with full integrated logistics support, including help desk, test bed, hardware maintenance, and depot support, to ensure the TSS system and its components meet the reliability, availability, and performance requirements in the *TSS System Specification*.
 9. **Operational support.** Provide training to TSS operators and on-call training support to TSS customers to enable the FAA to maximize training efficiency and effectiveness.
 10. **Remote software maintenance.** Implement TSS software updates remotely without interfering with training in progress or scheduled training.
 11. **VDB development and modification.** Accelerate the development, modification, and quality review of VDBs to meet user needs and be responsive to NAS operational requirements. Changes to a VDB must be completed and validated within 10 days of the request for changes. A new VDB representing an airport must be developed and validated within 30 days of the request.
 12. **Scenario development.** Accelerate the development, modification, and quality review of training scenarios to meet user needs and be responsive to NAS operational requirements. Development of a new training scenario or modifications to an existing training scenario must be completed and validated within 5 days of the request.
 13. **Operating efficiencies.** Realize operating cost efficiencies as compared to the current TSS system. The TSS solution must minimize the thermal and power load of TSS hardware to reduce the installation and operating costs associated with TSS training site air conditioning, power, and ventilation requirements.
 14. **Installation timeline.** Complete the installation of a TSS at every designated E2 training site, the development of a VDB and training scenarios for every ATCT, and the implementation of all elements of the TSS E2 solution by May 2028.

5. Constraints

The TSS E2 solution must align with FAA standards and orders, operate on commercial power, meet electromagnetic compatibility and noise requirements, and use commercial off-the-shelf (COTS) hardware components to the greatest extent possible.

The TSS E2 solution must adhere to FAA requirements for network connectivity, including security, compatibility, and performance requirements.

The TSS E2 solution must comply with relevant Federal laws regarding the procurement and use of hardware and telecommunications equipment.

6. Government-Furnished Equipment and Information

The FAA will provide a list of designated TSS training sites. The FAA will provide space, heating, cooling, and ventilation, power, and network connectivity at all TSS training sites.

The FAA will provide a list of VDBs. The TSS E2 vendor will have access to the FAA's existing library of VDBs and training scenarios.

The FAA will provide access to applicable FAA orders and policies.

7. Place of Performance

Work will occur at FAA facilities and remotely as applicable. The vendor will be expected to provide any facilities necessary to perform the depot/logistics, test bed, and offsite help desk functions.